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An end-to-end deep learning framework for accurate estimation of intracranial pressure waveform characteristics

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Abstract

It is clinically significant if intracranial pressure (ICP) waveform could be reliably reconstructed from simple physiological measurements that are commonly used in clinical practice. Some attempts have been made with the focus on estimating intermittent mean ICP (MICP) values, but no investigation has been performed to reconstruct beat-by-beat ICP waveforms from arterial blood pressure (ABP) waveforms. This study aimed

to overcome this challenge. A large dataset of 28511 segments (10.8 s for each segment; 540 samples) of simultaneously measured invasive ABP and ICP signals was used to develop and cross-evaluate a Wave-U-Net based deep learning model for reconstructing ICP waveform from ABP signals. From the reconstructed beat-by-beat ICP waveform, systolic ICP, diastolic ICP, and MICP were derived for each cardiac cycle and compared with reference ICPs. After comparing all the sampling points of each segment, the results showed that the reconstructed ICP waveforms were statistically correlated to the reference ICP waveforms, with the average Pearson's correlation coefficient (r) of 0.822 across all the segments. There was no significant difference between the estimated and reference ICP values, with the mean absolute error (MAE) of systolic ICP, diastolic ICP and MICP of 0.75 ± 0.29 , 0.48 ± 0.22 and 0.42 ± 0.18 mmHg, respectively, meeting the clinical standards of ICP monitoring for the management of traumatic brain injury. Our proposed method confirmed the hypothesis for the first time that ABP waveform can be used for reliable reconstruction of ICP waveform, enabling continuous monitoring of ICP waveform, and providing the hope possibility to replace the gold standard invasive technique.

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Keywords

- Intracranial pressure
- Arterial blood pressure
- Waveform estimation
- Deep learning

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KW - Deep learning

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